

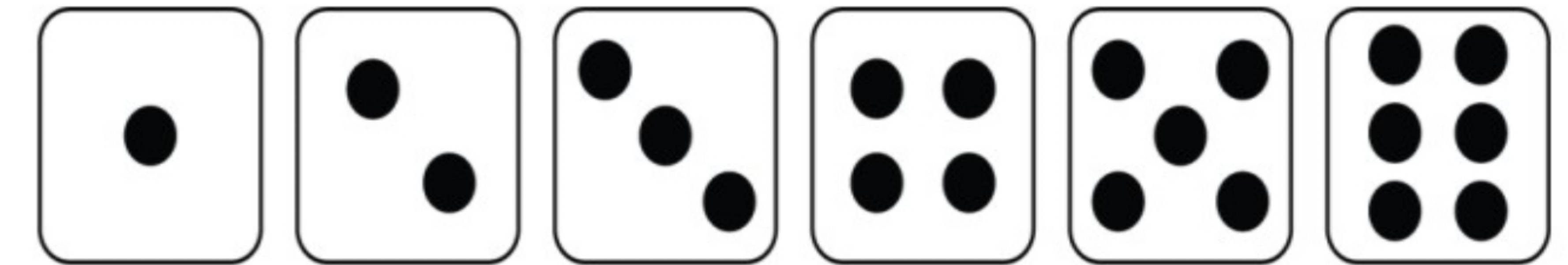
Probability

Sample space
for throwing
dice



i) For throwing a dice:-

$\{1, 2, 3, 4, 5, 6\}$



(ii) For throwing two dice :

exp



$S =$



Sample
Space

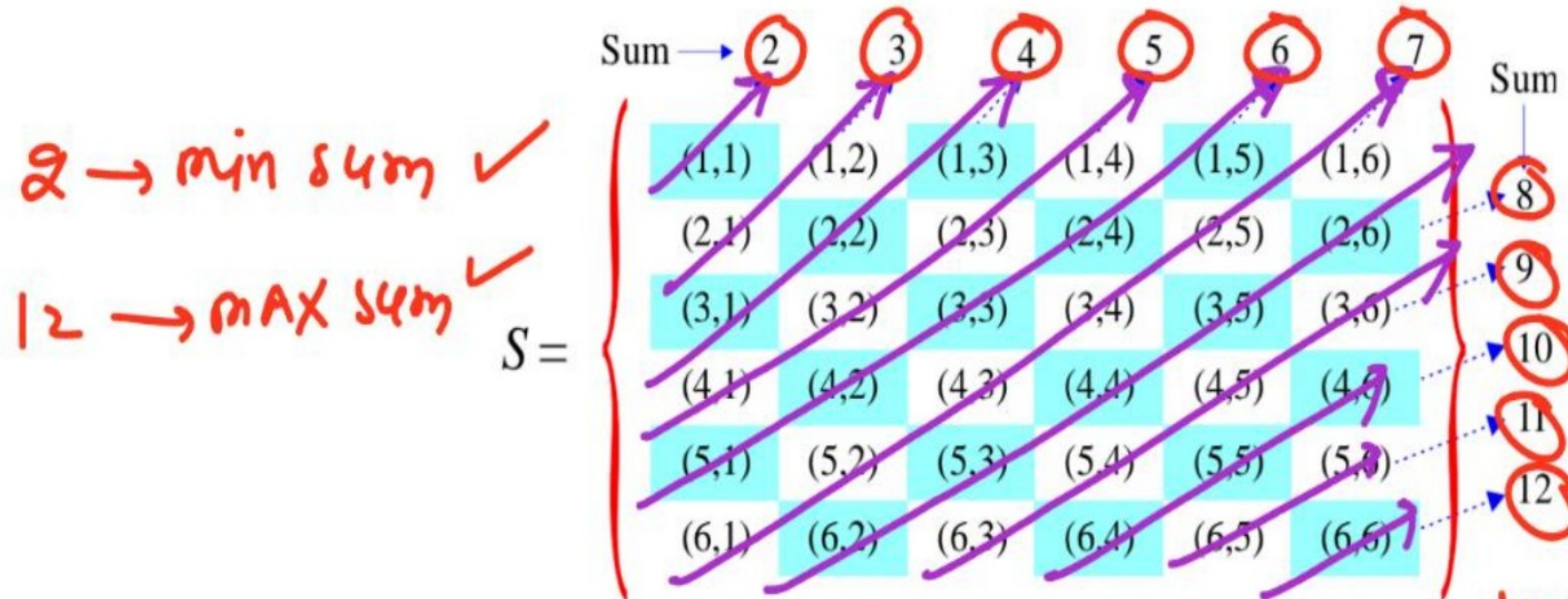
	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

6×6
 6×6
 6×6

6×6

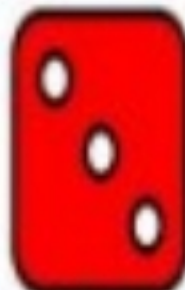
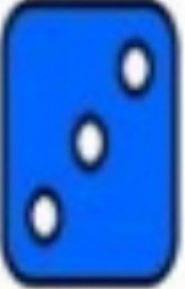
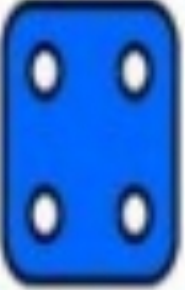
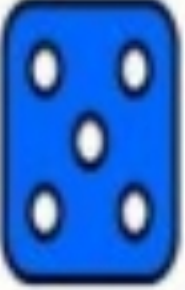
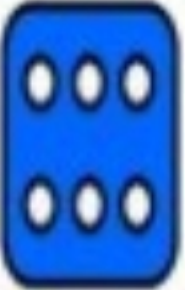
Probability

Sum of two dice (sample space and total favorable events)



Number of sample points = $36 = 6 \times 6 = 6^2$

Sample Space Diagrams

		Dice 1					
+							
Dice 2		2	3	4	5	6	7
		3	4	5	6	7	8
		4	5	6	7	8	9
		5	6	7	8	9	10
		6	7	8	9	10	11
		7	8	9	10	11	12
Total Score							

$$P(8) = \frac{5}{36}$$

$$P(9) = \frac{4}{36} = \frac{1}{9}$$

$$P(\text{even}) = \frac{18}{36} = \frac{1}{2}$$

$$P(0) = \frac{0}{36} = 0$$

$$P(\text{more than 7}) = \frac{15}{36} = \frac{5}{12}$$

$$P(3, 5 \text{ or } 10) = \frac{9}{36} = \frac{1}{4}$$

max sum



In general for throwing n dice :

$$S = 6^n \quad \checkmark$$

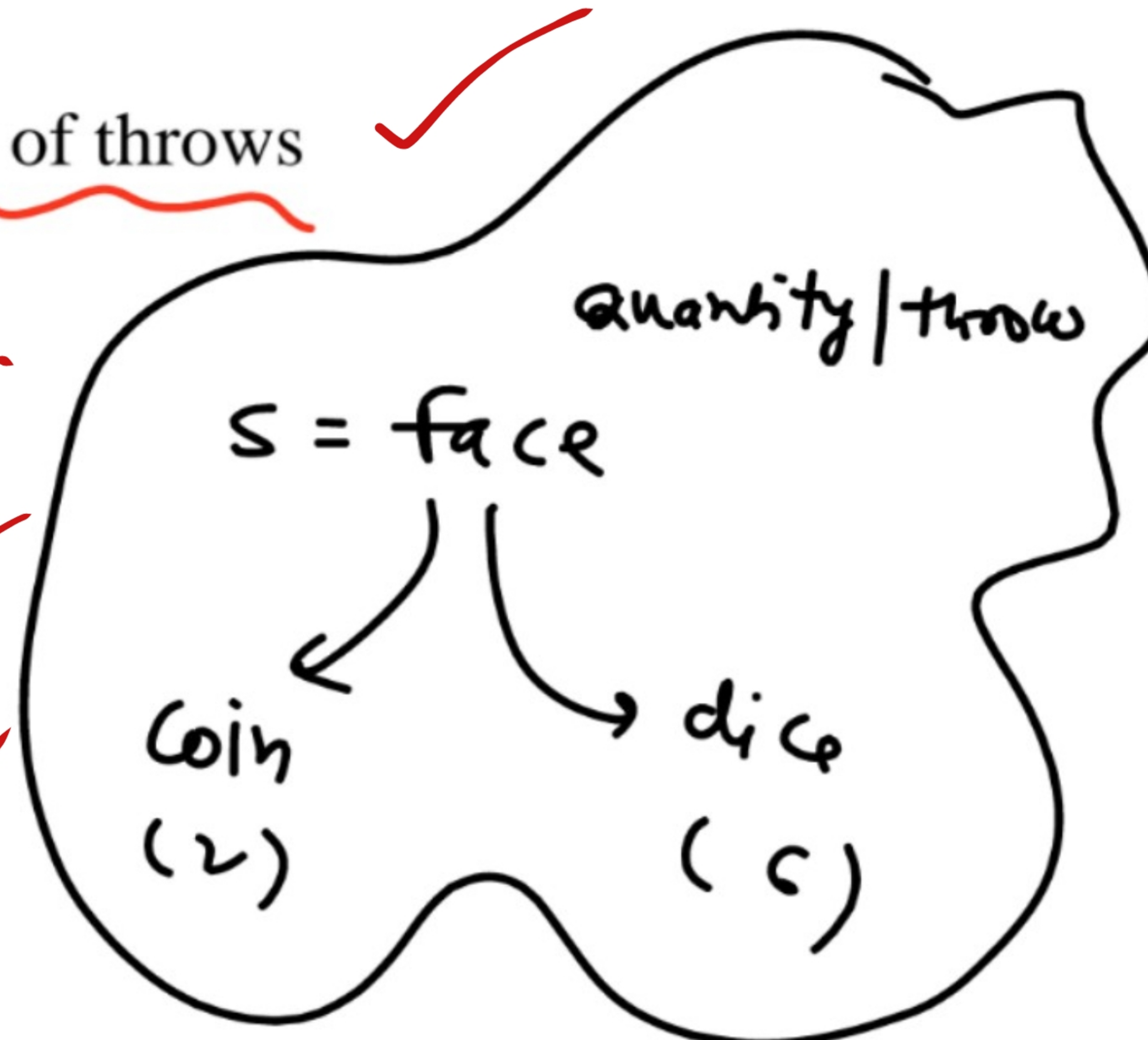
where, n = Number of dice or number of throws ✓

$$n = 1 \text{ die} \quad S = 6^1 = 6 \quad \checkmark$$

$$n = 2 \text{ dice} \quad S = 6^2 = 36 \quad \checkmark$$

$$n = 3 \text{ dice} \quad S = 6^3 = 216 \quad \checkmark$$

$$\begin{aligned} S &= 6^1 \\ S &= 6^2 \\ S &= 6^3 \end{aligned}$$



Probability

Triangle theory for dice :

When two dice are thrown then the possibility of getting total sum on the top surface of the dice is shown below,

Minimum sum of two dice = $[1,1] = 2$ ✓

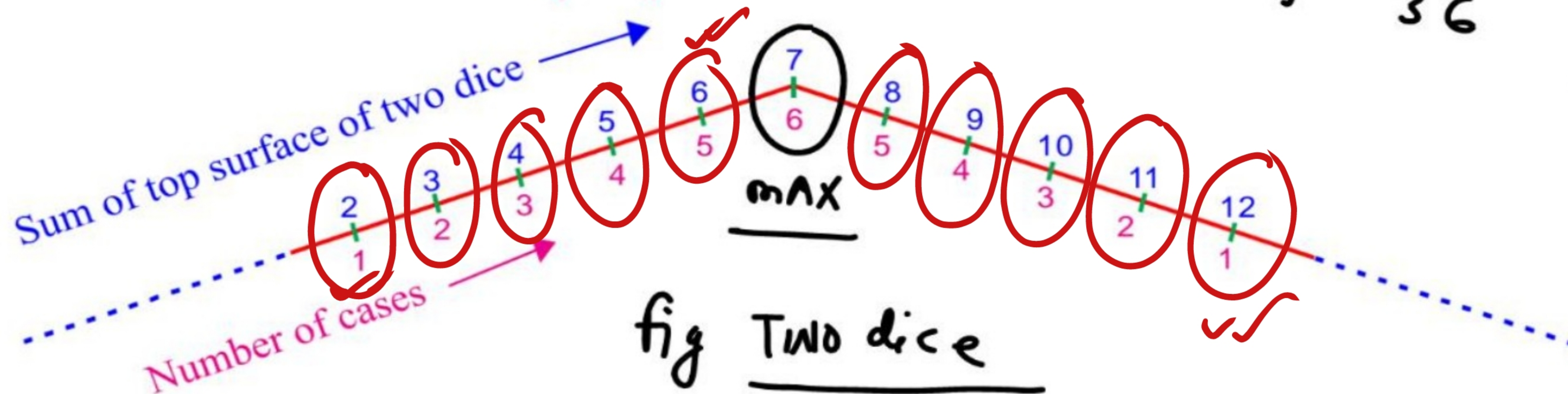
Maximum sum of two dice = $[6,6] = 12$ ✓

$$P[\text{sum} = 2] = \frac{1}{36}$$

$$P[\text{sum} = 12] = \frac{1}{36}$$

$$P[\text{sum} = 6] = \frac{5}{36}$$

(1,1)



❖ When a six-sided die rolls. What will be the probability of getting a prime number?

* what is the probability that the number is a perfect square? ✓

- (A) $\frac{1}{9}$
- (B) $\frac{1}{10}$
- (C) $\frac{9}{10}$
- (D) $\frac{1}{7}$
- (E) none

1, 4, 9, 16, 25, 36, 49, 64, 81, 100

$$P = \frac{\text{fav.}}{\text{total}} \\ = \frac{10}{100} = \frac{1}{10}$$



Prob:

Two coins are tossed 500 times.
and we get

Two heads: 105 times

One heads: 275 times

No head: 120 times

Find the probability of
each event to occur?

$$P(2H) = \frac{105}{500}$$

$$P(1H) = \frac{275}{500}$$

$$P(0H) = \frac{120}{500}$$



Prob.

Two dice are thrown at a time. Write down its Sample space.
Also find the probability that -

Sample space:

- (i) there will be a 'double six'. $(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$
- (ii) there will be a 'no six' and $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$
- (iii) there will be 'at least six' $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$
 $(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$
 ~~$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)$~~



Prob:

Two dice are thrown simultaneously (or, one die is thrown twice in succession)
find the probability that -

- ① The sum of the two results is $\rightarrow 10 \rightarrow \overset{(5,5) (6,4) (4,6)}{3/36 = 1/12}$
- ② the difference of the two results is 2,
- ③ The sum of the two results is 9 but their product is 20,
- ④ the sum of the two results is 14 -
- ⑤ the sum of the two results is divisible by 4 -

(ii) Difference = 2

↳ (1,3) (3,1) $\Rightarrow 8/36$
 (2,4) (4,2) $= 2/9$
 (3,5) (5,3)
 (4,6) (6,4)

(iii) Sum = 9 and product = 20

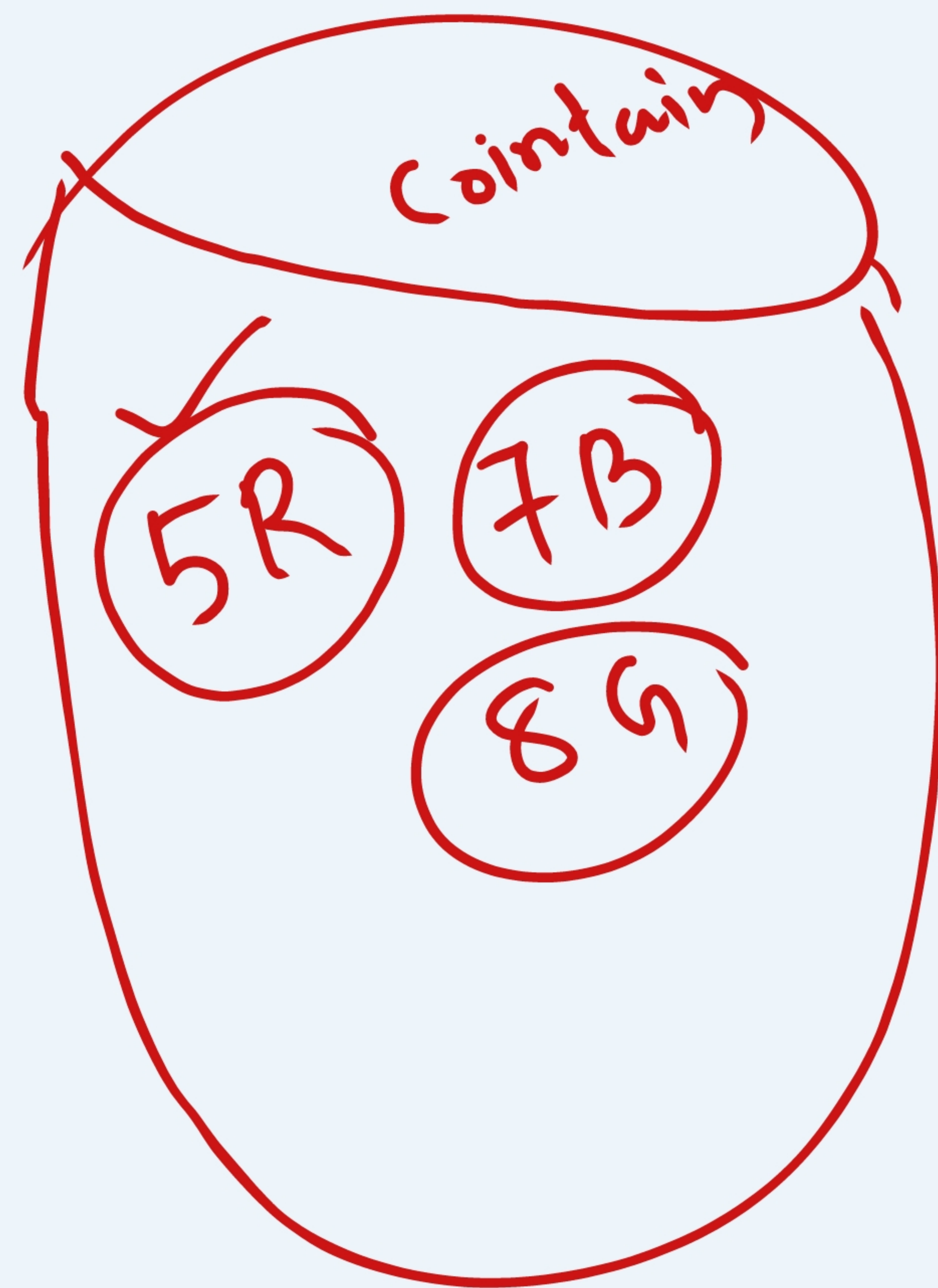
$\Rightarrow (5,4) (4,5)$
 $= 2/36 = 1/18$

(iv) Sum = 14

↳ Not possible = $0/36 = 0$

(v) Sum should be divisible by 4

↳ 4, 8, 12 $\Rightarrow \frac{3+5+1}{36}$
 ↙ ↘ ↘
 3 5 1 $\Rightarrow 1/9$



Bag

total no. balls = 20
 $\frac{5}{20} \times \frac{4}{19}$
 $\frac{1}{19}$

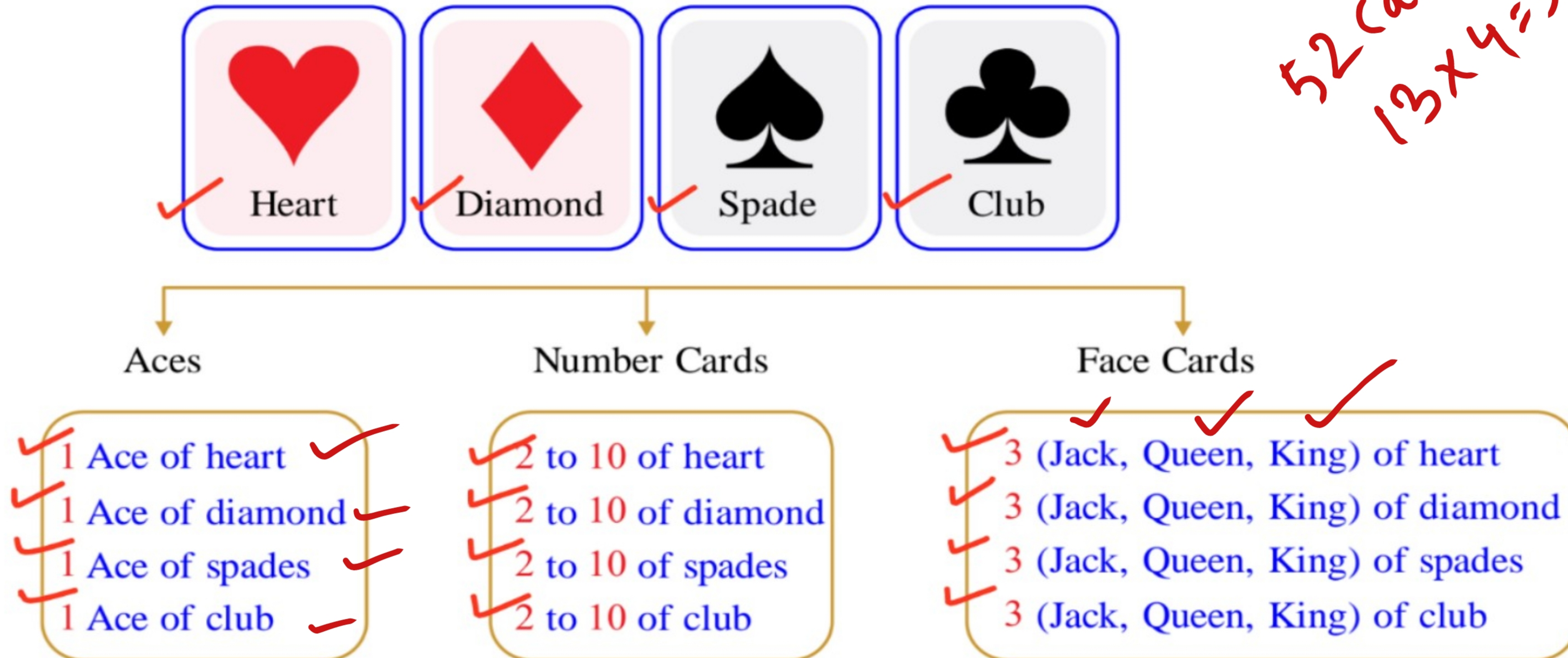
$$(20 - 1) = 19$$



2 balls are drawn
at random without
replacement. Find the Prob that both balls are
red?

Probability

Deck of Cards



Probability

Suits												Face Cards		
		Ace (A)	2	3	4	5	6	7	8	9	10	Jack	Queen	King
✓ Club		Non-Face Cards <u>$4 \times 10 = 40$ Cards</u>										✓	✓	✓
✓ Diamond												✓	✓	✓
✓ Heart												✓	✓	✓
✓ Spades												✓	✓	✓
												$4 \times 3 = 12$ Cards		



A card is drawn from the deck of cards. Find the probability of the following :

- (i) The probability of drawing a black card. $= \frac{26}{52}$
- (ii) The probability of drawing a heart shape card. $= \frac{13}{52}$
- (iii) The probability of drawing a face card. $= \frac{12}{52}$
- (iv) The probability of drawing a card with number 4. $= \frac{4}{52}$
- (v) The probability of drawing a red color card with number 4. $= \frac{2}{52}$

Find the probability of drawing a king from a deck of cards.

Find the probability of drawing a ace from a deck of cards.



Q. A card is drawn at random from a pack of well shuffled cards. what is the prob. that it is a heart or a queen?

$$\frac{N-1}{N} = \frac{13}{52} = \frac{1}{4}$$

$$13 + 3 = 16$$

$$\frac{16}{52} = \frac{4}{13}$$

$$\begin{aligned} P(H) &= \frac{13}{52} \\ P(Q) &= \frac{4}{52} \\ P(HQ) &= \frac{1}{52} \end{aligned}$$

$$\therefore P(H+Q) = P(H) + P(Q) - P(HQ)$$

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

or $P(A+B) = P(A) + P(B) - P(AB)$

Ans



Probability

probability of getting "♦" or 'Q' of ♥ ?
↓
Diamond
↓
Queen of heart

Applications of Probability in Data Science

1 Statistical inference

2 Machine learning

3 Bayesian analysis

4 Risk assessment

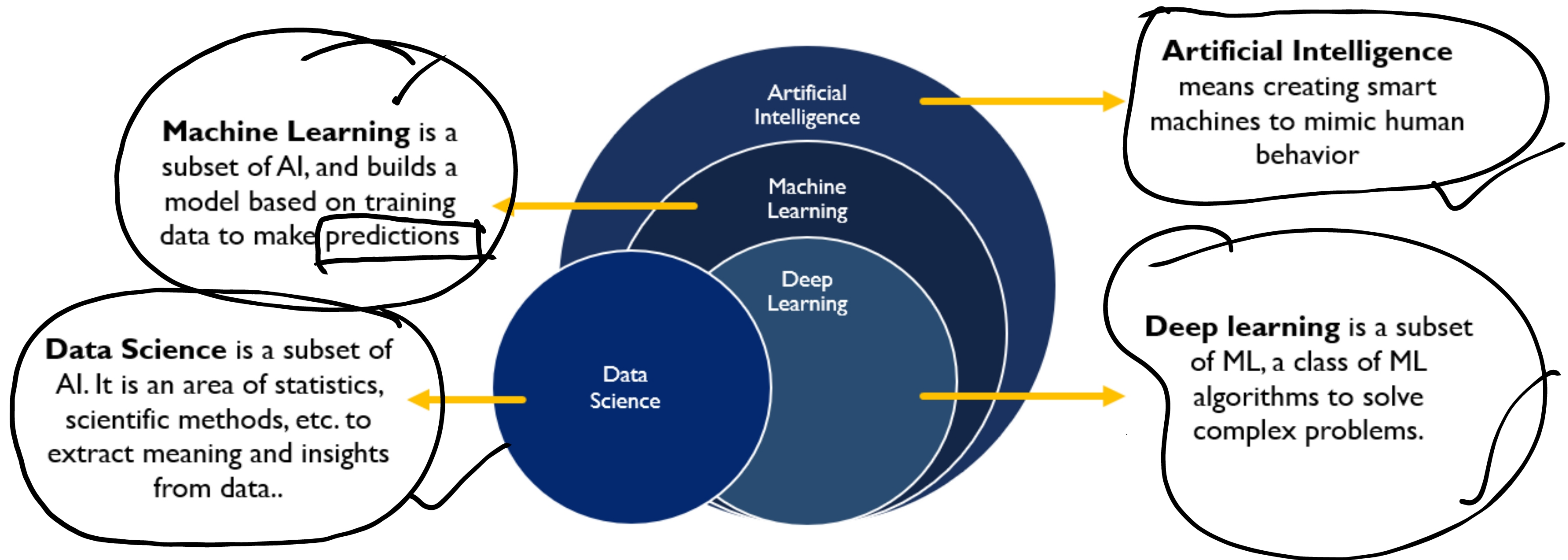
5 Quality control

6 Anomaly detection

Baye's
Theorem

Probability





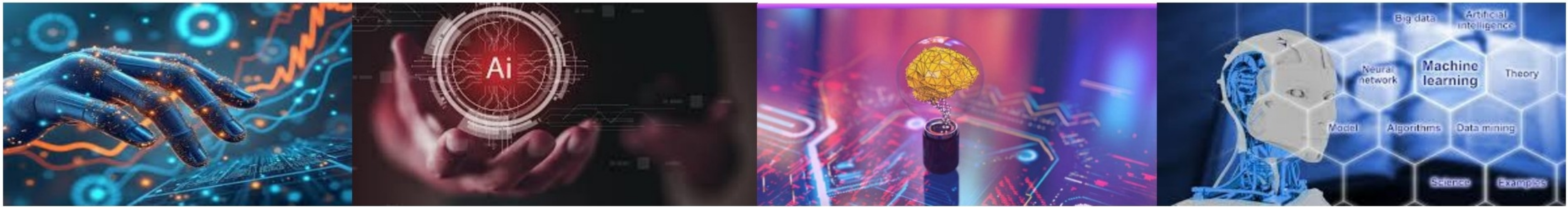
AI VS ML VS DL VS DATA SCIENCE



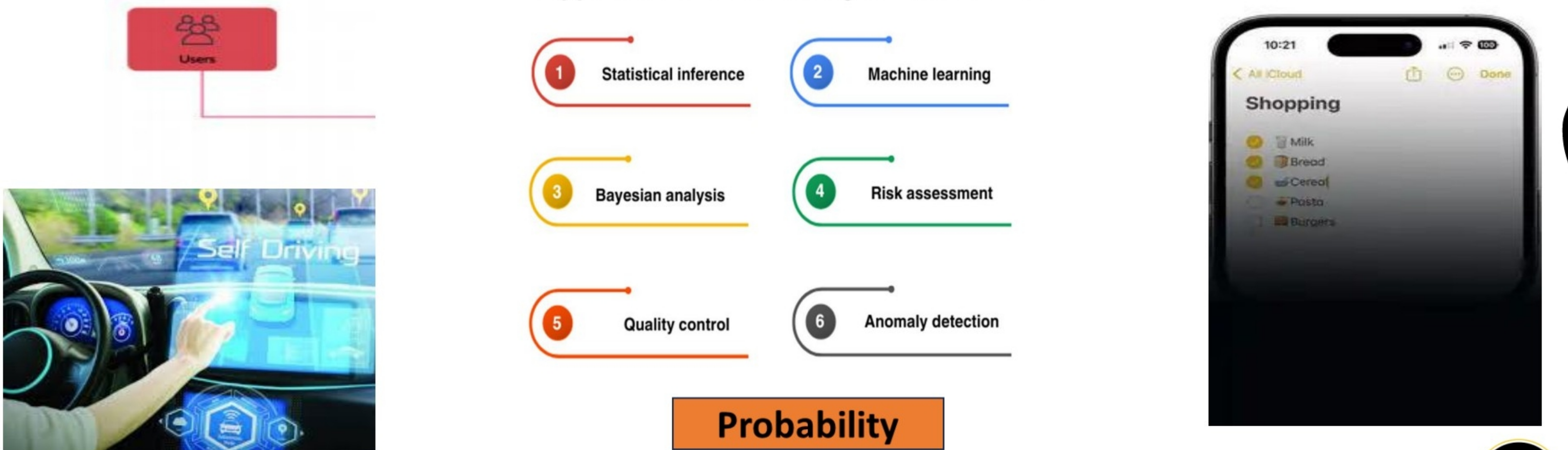
Quiz
10M (Monday)
① Eigenvalue & vectors

② SVD
③ C-H Theor.
④ Matrix Basics

Quad AI School of Technology



Applications of Probability in Data Science



1 Statistical inference 2 Machine learning

3 Bayesian analysis 4 Risk assessment

5 Quality control 6 Anomaly detection

Probability

Shopping

- Milk
- Bread
- Cereal
- Pasta
- Burgers

Mid-term Assessment

① Matrix
② Vector
③ Eigenvalue & vector
④ ~~Quiz~~
⑤ SVD
⑥ C-H Theor
⑦ Basics of Prob.